

DW_mult_seq

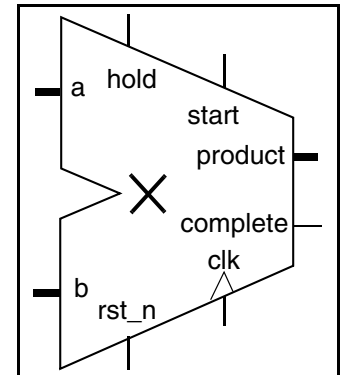
Sequential Multiplier

Version, STAR, and myDesignWare Subscriptions: [IP Directory](#)

Features and Benefits

- Parameterized word length
- Parameterized number of clock cycles
- Unsigned and signed (two's complement) data multiplication
- Registered or un-registered inputs and outputs.
- Includes a low-power implementation (at a sub-level) that has power benefits from minPower optimization (for details, see [Table 1-3](#) on page 2)

Revision History



Description

DW_mult_seq is a sequential multiplier designed for low area, area-time trade-off, or high frequency (small cycle time) applications.

Table 1-1 Pin Description

Pin Name	Width	Direction	Function
clk	1 bit	Input	Clock
rst_n	1 bit	Input	Reset, active low
hold	1 bit	Input	Hold current operation (=1)
start	1 bit	Input	Start operation (=1) A new operation is started again by making <code>start = 1</code> for one clock cycle.
a	<i>a_width</i> bits	Input	Multiplier
b	<i>b_width</i> bits	Input	Multiplicand
complete	1 bit	Output	Operation completed (=1)
product	<i>a_width</i> + <i>b_width</i> bits	Output	Product $a \times b$

Table 1-2 Parameter Description

Parameter	Values	Description
a_width	≥ 3 and $\leq b_width$	Word length of a
b_width	≥ 3	Word length of b
tc_mode	0 or 1 Default: 0	Two's complement control 0: Unsigned 1: Two's complement
num_cyc	≥ 3 and $\leq a_width$ Default: 3	User-defined number of clock cycles to produce a valid result The real number of clock cycles depends on various parameters and is given in Table 1-6 on page 4 and the topic titled “ Formula for Multiplier Bits Processed Per Cycle ” on page 4.
rst_mode	0 or 1 Default: 0	Reset mode 0: Asynchronous reset 1: Synchronous reset
input_mode ^a	0 or 1 Default: 1	Registered inputs 0: No 1: Yes
output_mode	0 or 1 Default: 1	Registered outputs 0: No 1: Yes
early_start	0 or 1 Default: 0	Computation start 0: Start computation in the second cycle 1: Start computation in the first cycle See Table 1-6 on page 4 for the dependency of <i>early_start</i> on <i>input_mode</i> .

a. When configured with the parameter *input_mode* = 0, the inputs a and b MUST be held constant from the time *start* is asserted until *complete* has gone high to signal completion of the calculation. Conversely, if a configuration with the parameter *input_mode* = 1 is used, the a and b inputs will be captured when *start* is high and otherwise ignored.

Table 1-3 Synthesis Implementations

Implementation	Function	License Feature Required
cpa ^a	Carry-propagate adder synthesis model	DesignWare ^b

a. To achieve low-power benefits in sub-module implementations, you need to enable minPower; for details, see “[Enabling minPower](#)” on page 8.

b. For releases prior to P-2019.03, the DesignWare-LP license feature is required to achieve low-power benefits.

Table 1-4 Simulation Models

Model ^a	Function
DW03.DW_MULT_SEQ_CFG_SIM	Design unit name for VHDL simulation
dw/dw03/src/DW_mult_seq_sim.vhd	VHDL simulation model source code
dw/sim_ver/DW_mult_seq.v	Verilog simulation model source code

a. Note that during the computation phase (after start and before complete is asserted), the simulation models output X values and therefore cannot be used as a compare for gate-level simulations.

Table 1-5 Operation Truth Table

start	hold	next state
0	0	Idle or Running
0	1	Hold
1	X	Start

DW_mult_seq multiplies the operand *a* by *b* to produce a result (*product*) in a user-defined number of clock cycles (*num_cyc*). As long as *start* = 1, the multiplication operation is in the initialization state. Once *start* = 0, the calculation begins followed by valid output flagged when *complete* = 1. The multiply operation is stalled when *hold* is 1.

The parameter *tc_mode* determines whether the data of inputs (*a*, *b*) and output (*product*) is interpreted as unsigned (*tc_mode* = 0) or two's complement (*tc_mode* = 1) numbers.

The internal registers can either have an asynchronous (*rst_mode* = 0) or synchronous reset (*rst_mode* = 1) that is connected to the reset signal *rst_n*.

Within the first *num_cyc* clock cycles immediately after reset conditions are released (*rst_n* = 1), *start* must remain 0 until the first assertion of *complete* (that is, *complete* = 1). This first *complete* = 1 following reset may yield invalid results and should be disregarded.

The parameter *input_mode* determines whether the inputs are to be registered inside DW_mult_seq (*input_mode* = 1) or not (*input_mode* = 0). If configured without input registers (*input_mode* = 0), then the logic that drives the inputs *a* and *b* must hold the input values constant for the entire time it takes to calculate the result (from the cycle before *start* drops until *complete* goes high). When configured with input registers (*input_mode* = 1) the inputs *a* and *b* are captured when *start* is high and ignored until *start* goes high again.



When configured with no input registers, changes on inputs *a* and *b* while *complete* is low (calculation cycle) will produce unpredictable output values. Simulation models will produce unknown output values (Xs) and post an error message indicating the instance that violated this rule and the simulation time when the violation was detected.

The parameter *output_mode* determines whether the outputs are registered (*output_mode* = 1) or not (*output_mode* = 0).

When parameter *early_start* = 1, computation starts immediately after setting the *start* to 1. This saves one extra cycle to store the data (*early_start* = 0), but feeds the inputs directly into the components critical path. Table 1-6 on page 4 shows the *input_mode*, *output_mode*, and *early_start* parameter combinations and corresponding actual number of cycles required to perform an operation.

Table 1-6 Actual Cycles Based on *input_mode*, *output_mode*, and *early_start*

<i>input_mode</i>	<i>output_mode</i>	<i>early_start</i>	Actual Number of Cycles
0	0	0	<i>num_cyc</i> -2
0	0	1	Invalid parameter setting
0	1	0	<i>num_cyc</i> -1
0	1	1	Invalid parameter setting
1	0	0	<i>num_cyc</i> -1
1	0	1	<i>num_cyc</i> -2
1	1	0	<i>num_cyc</i>
1	1	1	<i>num_cyc</i> -1

Note that the *num_cyc* value indicates the actual throughput of the device from when *start* is asserted to when *complete* is asserted. However, if a calculation is in progress (before the *num_cyc* number of cycles has been reached) when *start* is asserted again, the results are undetermined until *complete* is asserted. The results associated with the assertion of *complete* are from the input values from the previous assertion of *start*.

Formula for Multiplier Bits Processed Per Cycle

The following formula describes the number of multiplier bits processed per cycle:

$$\text{bits processed per cycle} = \text{ceil} (a_width / num_cyc)$$

where:

a_width is the bit width of the multiplier (as defined in Table 1-2 on page 2)

num_cyc is the number of clock cycles required for multiplication (as defined in Table 1-2 on page 2)

Formula for Actual Number of Cycles Required

The actual number of clock cycles required for a computation is calculated using the following formula:

$$\text{actual number of cycles required} = num_cyc - (1 - output_mode) - (1 - input_mode) - early_start$$

where:

num_cyc is the number of clock cycles required for division (as defined in Table 1-2 on page 2)

output_mode is the control for registered output (as defined in Table 1-2 on page 2)

input_mode is the control for registered inputs (as defined in Table 1-2 on page 2)

early_start is the control for when the computation starts (as defined in Table 1-2 on page 2)

Suppressing Warning Messages During Verilog Simulation

The Verilog simulation model includes macros that allow you to suppress warning messages during simulation.

To suppress all warning messages for all DWBB components, define the DW_SUPPRESS_WARN macro in either of the following ways:

- Specify the Verilog preprocessing macro in Verilog code:

```
`define DW_SUPPRESS_WARN
```

- Or, include a command line option to the simulator, such as:

```
+define+DW_SUPPRESS_WARN (which is used for the Synopsys VCS simulator)
```

The warning messages for this model include the following:

- If values other than 1 or 0 are present on a clock port, the following message is displayed:

```
WARNING: <instance_path>.<clock_name>_monitor:  
at time = <timestamp>, Detected unknown value, x, on <clock_name> input.
```

To suppress only this warning message for all DWBB components, use the following macro:

- Define the DW_DISABLE_CLK_MONITOR macro. You can define this macro in the following ways:

- Specify the Verilog preprocessing macro in Verilog code:

```
`define DW_DISABLE_CLK_MONITOR
```

- Or, include a command line option to the simulator, such as:

```
+define+DW_DISABLE_CLK_MONITOR (which is used for the Synopsys VCS simulator)
```

This message is also suppressed using the DW_SUPPRESS_WARN macro explained earlier.

- If the component is configured without an input register and an input operand changes during calculation, the following message is displayed:

```
WARNING: <instance_path>:  
at time = <timestamp>, Operand input change on DW_mult_seq during calculation  
(configured without an input register) will cause corrupted results if operation is  
allowed to complete.
```

To suppress this message, use the DW_SUPPRESS_WARN macro explained earlier.

- If the component is configured without input and output registers and an input operand changes during calculation, the following message is displayed:

```
WARNING: <instance_path>:  
at time = <timestamp>, Operand input change on DW_mult_seq during calculation  
(configured with neither input nor output register) causes output to no longer retain  
result of previous operation.
```

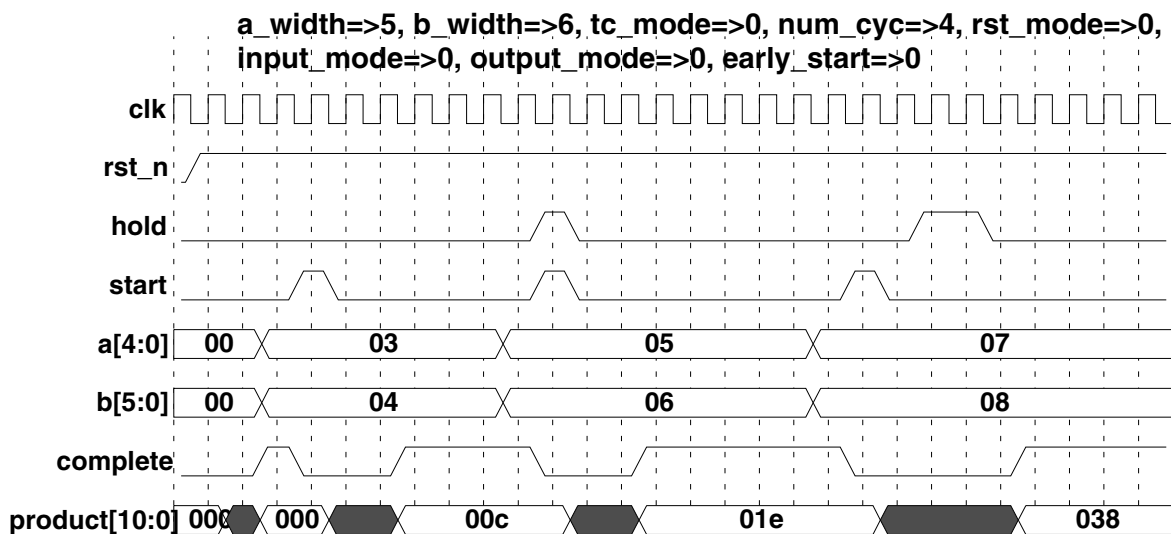
To suppress this message, use the DW_SUPPRESS_WARN macro explained earlier.

Timing Waveforms

The following timing waveforms show a 5-bit by 6-bit unsigned sequential multiplier for specific inputs of `hold` and `start` and their corresponding outputs. The parameter settings for each simulation are shown at the top of each figure. When `hold = 1` and `start = 0`, the result is delayed by the same number of clock cycles for which `hold` is 1. For example, if `hold = 1` for two clock cycles, the result is delayed by two clock cycles.

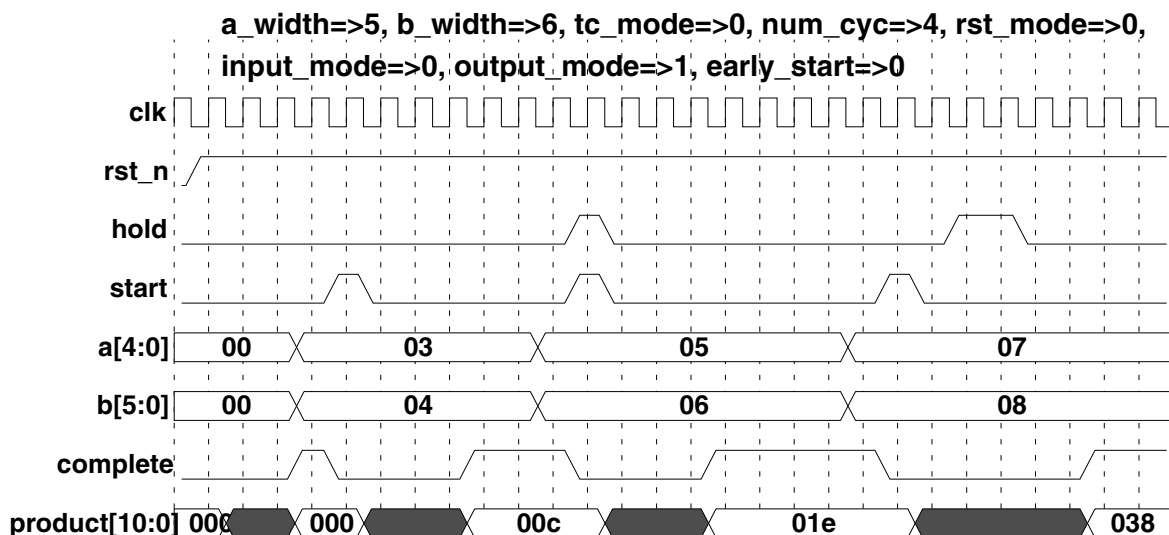
For the parameter settings shown in [Figure 1-1](#), [Table 1-6](#) on page 4 specifies that the result is produced after two cycles. However, the data is available on the fourth edge following the assertion of the `start` signal.

Figure 1-1 Simulation Waveform 1



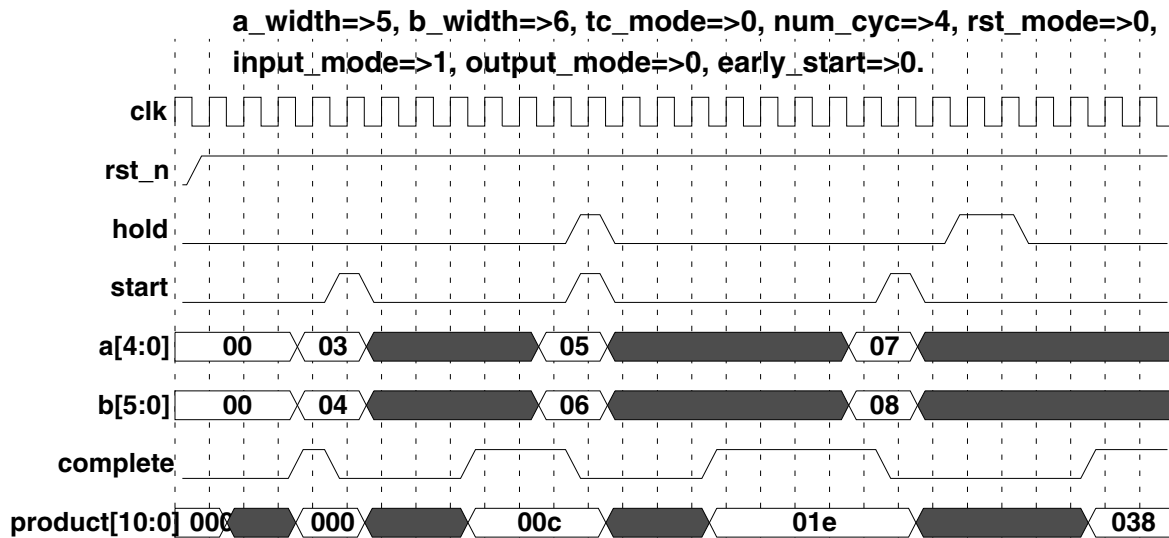
For the parameter settings shown in [Figure 1-2](#), [Table 1-6](#) on page 4 specifies that the result is produced after three cycles.

Figure 1-2 Simulation Waveform 2



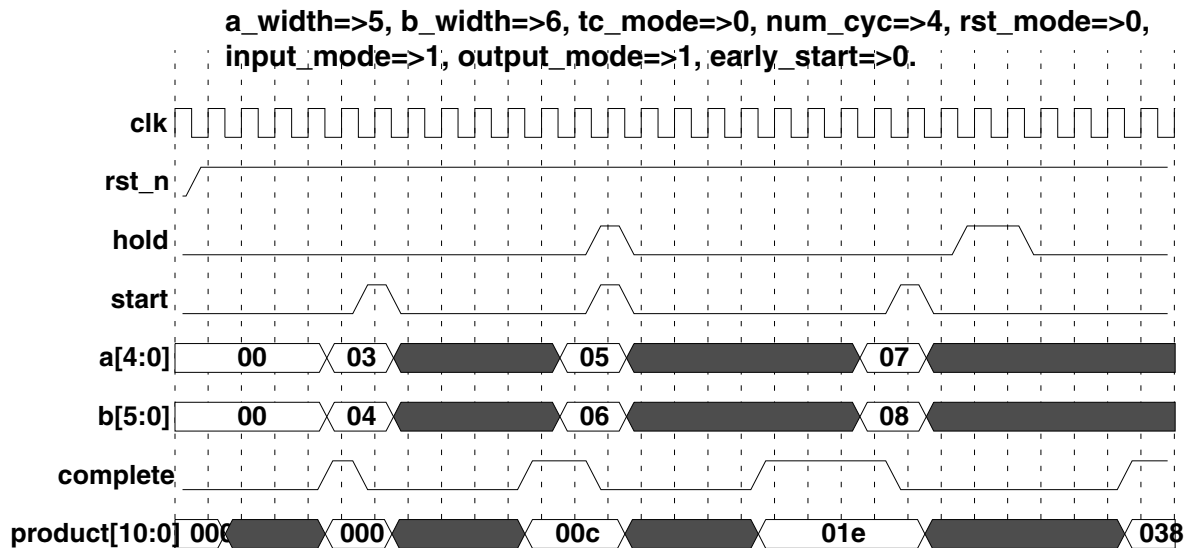
For the parameter settings shown in [Figure 1-3](#), [Table 1-6](#) on page 4 specifies that the result is produced after three cycles. Since *input_mode* = 1 (registered input) the input data can be removed after the first cycle.

Figure 1-3 Simulation Waveform 3



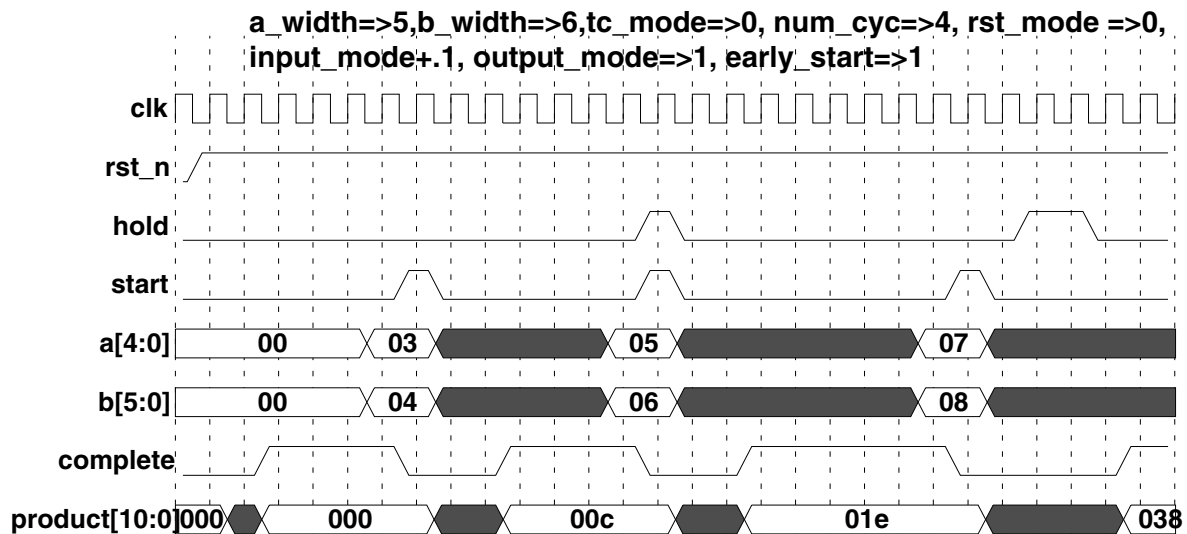
For the parameter settings shown in [Figure 1-4](#), [Table 1-6](#) on page 4 specifies that the result is produced after four cycles. Since *input_mode* = 1 (registered input), the input data can be removed after the first cycle.

Figure 1-4 Simulation Waveform 4



For the parameter settings shown in [Figure 1-5](#), [Table 1-6](#) on page 4 specifies that the result is produced after three clock cycles. Because *input_mode* = 1 (registered input), the input data can be removed after the first cycle. With *hold* = 1 and *start* = 1, the result is delayed by the same number of cycles that *hold* = 1. Note that the data will be available on the *num_cyc* number of clocks after the data is registered. Note that the data is registered in the clock cycle immediately proceeding *start* = 1.

Figure 1-5 Simulation Waveform 5



Enabling minPower

You can instantiate this component without enabling minPower, but to achieve power savings from the low-power implementation (at a sub-level--see [Table 1-3](#) on page 2), you must enable minPower optimization, as follows:

- Design Compiler
 - Version P-2019.03 and later:


```
set power_enable_minpower true
```
 - Before version P-2019.03 (requires the DesignWare-LP license feature):


```
set synthetic_library {dw_foundation.sldb dw_minpower.sldb}
set link_library {* $target_library $synthetic_library}
```

- Fusion Compiler

Optimization for minPower is enabled as part of the *total_power* metric setting. To enable the *total_power* metric, use the following:

```
set_qor_strategy -stage synthesis -metric total_power
```

Related Topics

- [Math – Sequential Overview](#)
- [DesignWare Building Block IP User Guide](#)

HDL Usage Through Component Instantiation - VHDL

```

library IEEE, DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DW_Foundation_comp_arith.all;

entity DW_mult_seq_inst is
  generic (inst_a_width      : POSITIVE := 8; inst_b_width      : POSITIVE := 8;
           inst_tc_mode      : INTEGER := 0; inst_num_cyc      : INTEGER := 3;
           inst_rst_mode     : INTEGER := 0; inst_input_mode    : INTEGER := 1;
           inst_output_mode  : INTEGER := 1; inst_early_start   : INTEGER := 0
          );
  port (inst_clk : in std_logic;  inst_rst_n : in std_logic;
        inst_hold : in std_logic; inst_start : in std_logic;
        inst_a : in std_logic_vector(inst_a_width-1 downto 0);
        inst_b : in std_logic_vector(inst_b_width-1 downto 0);
        complete_inst : out std_logic;
        product_inst : out
                      std_logic_vector(inst_a_width+inst_b_width-1 downto 0)
        );
end DW_mult_seq_inst;

architecture inst of DW_mult_seq_inst is
begin
  -- Instance of DW_mult_seq
  U1 : DW_mult_seq
    generic map (a_width => inst_a_width,  b_width => inst_b_width,
                 tc_mode => inst_tc_mode,   num_cyc => inst_num_cyc,
                 rst_mode => inst_rst_mode,  input_mode => inst_input_mode,
                 output_mode => inst_output_mode,
                 early_start => inst_early_start
                )
    port map (clk => inst_clk,  rst_n => inst_rst_n,  hold => inst_hold,
              start => inst_start,  a => inst_a,  b => inst_b,
              complete => complete_inst,  product => product_inst
             );
end inst;

-- pragma translate_off
configuration DW_mult_seq_inst_cfg_inst of DW_mult_seq_inst is
  for inst
  end for;
end DW_mult_seq_inst_cfg_inst;
-- pragma translate_on

```

HDL Usage Through Component Instantiation - Verilog

```
module DW_mult_seq_inst(inst_clk, inst_rst_n, inst_hold, inst_start, inst_a,
                       inst_b, complete_inst, product_inst );
    parameter inst_a_width = 8;
    parameter inst_b_width = 8;
    parameter inst_tc_mode = 0;
    parameter inst_num_cyc = 3;
    parameter inst_rst_mode = 0;
    parameter inst_input_mode = 1;
    parameter inst_output_mode = 1;
    parameter inst_early_start = 0;
    // Please add +incdir+$SYNOPSYS/dw/sim_ver+ to your verilog simulator
    // command line (for simulation).
    input inst_clk;
    input inst_rst_n;
    input inst_hold;
    input inst_start;
    input [inst_a_width-1 : 0] inst_a;
    input [inst_b_width-1 : 0] inst_b;
    output complete_inst;
    output [inst_a_width+inst_b_width-1 : 0] product_inst;
    // Instance of DW_mult_seq
    DW_mult_seq #(inst_a_width, inst_b_width, inst_tc_mode, inst_num_cyc,
                  inst_rst_mode, inst_input_mode, inst_output_mode,
                  inst_early_start)
    U1 (.clk(inst_clk), .rst_n(inst_rst_n), .hold(inst_hold),
        .start(inst_start), .a(inst_a), .b(inst_b),
        .complete(complete_inst), .product(product_inst) );
endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

Date	Release	Updates
September 2024	DWBB_202409.0	Updated date and version; no technical changes
December 2023	DWBB_202312.0	Updated date and version; no technical changes
January 2023	DWBB_202212.1	- Clarified the note in Table 1-6 on page 4 - Clarified formulas in “Formula for Multiplier Bits Processed Per Cycle” on page 4 and “Formula for Actual Number of Cycles Required” on page 4
July 2020	DWBB_201912.5	Adjusted content and title of “Suppressing Warning Messages During Verilog Simulation” on page 5 and added the DW_SUPPRESS_WARN macro
October 2019	DWBB_201903.5	Added the “Disabling Clock Monitor Messages” section
March 2019	DWBB_201903.0	- Clarified license requirements in Table 1-3 on page 2 - Added “Enabling minPower” on page 8 - Added this Revision History table and the document links on this page

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